

DEPARTMENT:PDI

LAB 05

Course: PDI

Group: I2-IP-Group AIECS

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Institute of Technology of Cambodia

Practice 1

Exercise1.java

Exercise1.java | Exercise2.java | Book.java | Main.java | Employee.java

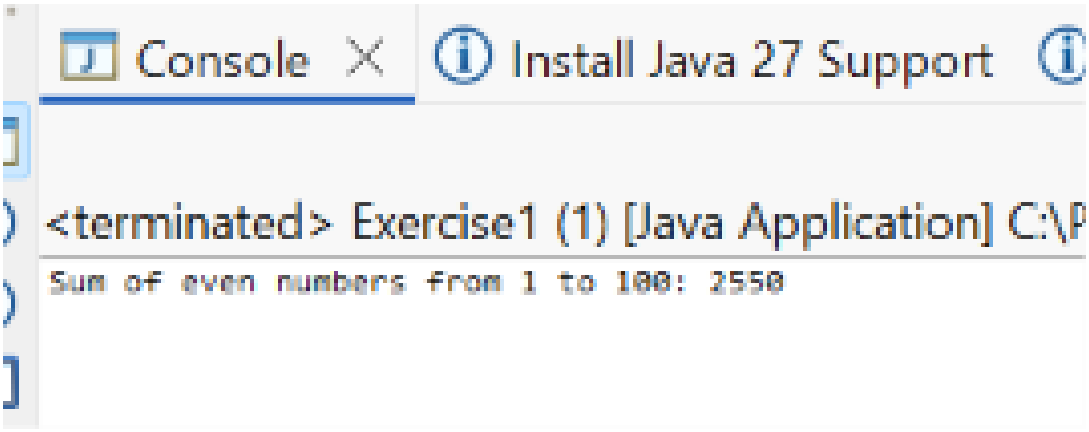
```java
package Exercise1;
public class Exercise1 {
    public static void main(String[] args) {

        int sum = 0;

        for (int i = 1; i <= 100; i++) {
            if (i % 2 == 0) {
                sum = sum + i;
            }
        }

        System.out.println("Sum of even numbers from 1 to 100: " + sum);
    }
}
```

Output:



Practice 2

Exercise2.java:

Exercise1.java | Exercise2.java | Book.java

```java
package Exercise2;
import java.util.Scanner;

public class Exercise2 {
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.println("===== CALCULATOR =====");
        System.out.println("1. Addition (+)");
        System.out.println("2. Subtraction (-)");
        System.out.println("3. Multiplication (*)");
        System.out.println("4. Division (/)");
        System.out.println("5. Sum from 1 to n");

        System.out.print("Choose an option: ");
        int choice = input.nextInt();

        switch (choice) {

            case 1:
                System.out.print("Enter first number: ");
                double a = input.nextDouble();

                System.out.print("Enter second number: ");
                double b = input.nextDouble();

                System.out.println("Result: " + (a + b));
                break;

            case 2:
                System.out.print("Enter first number: ");
                double c = input.nextDouble();

                System.out.print("Enter second number: ");
                double d = input.nextDouble();

                System.out.println("Result: " + (c - d));
                break;

            case 3:
                System.out.print("Enter first number: ");
                double e = input.nextDouble();

                System.out.print("Enter second number: ");
                double f = input.nextDouble();

                System.out.println("Result: " + (e * f));
                break;

            case 4:
                System.out.print("Enter first number: ");
                double x = input.nextDouble();

                System.out.print("Enter second number: ");
                double y = input.nextDouble();

                if (y != 0) {
                    System.out.println("Result: " + (x / y));
                } else {
                    System.out.println("Error: Cannot divide by zero.");
                }
                break;

            case 5:
                System.out.print("Enter n: ");
                int n = input.nextInt();

                int sum = 0;

                for (int i = 1; i <= n; i++) {
                    sum += i;
                }

                System.out.println("Sum from 1 to " + n + " = " + sum);
                break;

            default:
                System.out.println("Invalid option.");
        }

        input.close();
    }
}
```

Output:

Console | Install Java

<terminated> Exercise2 (1) [Java

```
===== CALCULATOR =====
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Sum from 1 to n
Choose an option: 1
Enter first number: 3
Enter second number: 23
Result: 26.0
```

Practice 3

Book.java

Exercise1.java | Exercise2.java | Book.java

```java
package Exercise3;


public class Book {

    String title;
    String author;
    double price;

    public Book() {
        title = "Unknown";
        author = "Unknown";
        price = 0.0;
    }
    public Book(String title, String author, double price) {
        this.title = title;
        this.author = author;
        this.price = price;
    }
    public void displayDetails() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: $" + price);
    }
}

```

Main.java

```java
package Exercise3;


public class Main {
    public static void main(String[] args) {


        Book book1 = new Book();


        Book book2 = new Book(
            "Java Programming",
            "James Gosling",
            25.50
        );

        System.out.println("Book 1:");
        book1.displayDetails();

        System.out.println();

        System.out.println("Book 2:");
        book2.displayDetails();
    }
}

```

Output:

Console | Install Java 27 Su

<terminated> Main (3) [Java Applicatic

```
Book 1:
Title: Unknown
Author: Unknown
Price: $0.0

Book 2:
Title: Java Programming
Author: James Gosling
Price: $25.5
```

Practice4

Employee.java

Exercise1.java Exercise2.java Book.java

```java
package Exercise4;


public class Employee {

    private int id;
    private String name;
    private int age;
    private String department;
    private double salary;
    public Employee(int id, String name, int age,
                    String department, double salary) {

        this.id = id;
        this.name = name;
        this.age = age;
        this.department = department;
        this.salary = salary;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public int getAge() {
        return age;
    }

    public String getDepartment() {
        return department;
    }

    public double getSalary() {
        return salary;
    }

    // Setters
    public void setId(int id) {
        this.id = id;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setAge(int age) {
        this.age = age;
    }

    public void setDepartment(String department) {
        this.department = department;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }

    public void displayInfo() {
        System.out.println("Employee ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Department: " + department);
        System.out.println("Salary: $" + salary);
    }
}
```

Main:

Exercise1.java Exercise2.java Book.java Main

```java
package Exercise4;

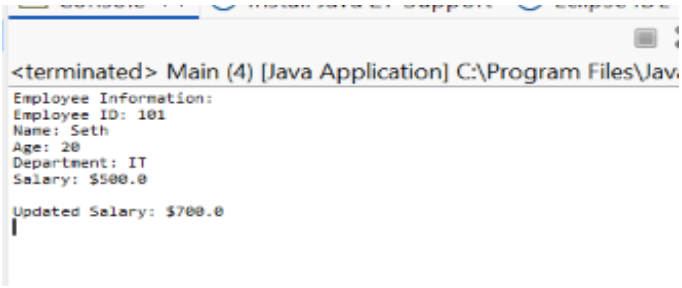
Week5/src/Exercise1/Exercise1.java

public class Main {
    public static void main(String[] args) {

        Employee emp = new Employee(
            101, "Seth", 20, "IT", 500
        );

        System.out.println("Employee Information:");
        emp.displayInfo();
        emp.setSalary(700);

        System.out.println("\nUpdated Salary: $" + emp.getSalary());
    }
}
```



Output:

Practice 5:

Exercise5.java

```java
package Exercise5;
import java.math.BigInteger;
public class Exercise5 {
    public static BigInteger factorialLoop(int n) {
        BigInteger result = BigInteger.ONE;
        for (int i = 1; i <= n; i++) {
            result = result.multiply(BigInteger.valueOf(i));
        }
        return result;
    }
    public static BigInteger factorialRecursive(int n) {

        if (n == 0 || n == 1) {
            return BigInteger.ONE;
        }

        return BigInteger.valueOf(n)
                .multiply(factorialRecursive(n - 1));
    }

    public static void main(String[] args) {

        int[] numbers = {3, 6, 10, 50};

        for (int n : numbers) {

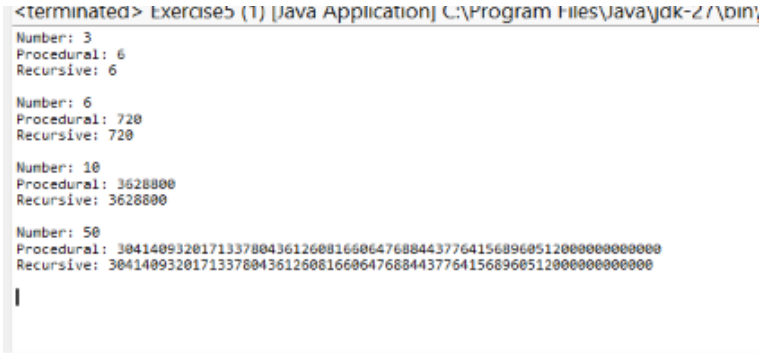
            System.out.println("Number: " + n);

            System.out.println(
                "Procedural: " + factorialLoop(n)
            );

            System.out.println(
                "Recursive: " + factorialRecursive(n)
            );

            System.out.println();
        }
    }
}
```

Output:



Practice 6:

Ractangle.java

```java
package Exercise6;

public class Rectangle {

    private double width;
    private double height;

    public Rectangle(double width, double height) {
        this.width = width;
        this.height = height;
    }

    public double getWidth() {
        return width;
    }

    public void setWidth(double width) {
        this.width = width;
    }

    public double getHeight() {
        return height;
    }

    public void setHeight(double height) {
        this.height = height;
    }

    public double calculateArea() {
        return width * height;
    }

    public double calculatePerimeter() {
        return 2 * (width + height);
    }
}
```

Circle.java

```java
package Exercise6;


public class Circle {

    private double radius;

    public Circle(double radius) {
        this.radius = radius;
    }

    public double getRadius() {
        return radius;
    }

    public void setRadius(double radius) {
        this.radius = radius;
    }

    public double calculateArea() {
        return Math.PI * radius * radius;
    }

    public double calculatePerimeter() {
        return 2 * Math.PI * radius;
    }
}
```

Main.java

```java
package Exercise6;

public class Main {
    public static void main(String[] args) {

        Rectangle rectangle = new Rectangle(5, 4);

        System.out.println("RECTANGLE");
        System.out.println("Width: " + rectangle.getWidth());
        System.out.println("Height: " + rectangle.getHeight());
        System.out.println("Area: " + rectangle.calculateArea());
        System.out.println("Perimeter: " +
                           rectangle.calculatePerimeter());

        Circle circle = new Circle(3);

        System.out.println("\nCIRCLE");
        System.out.println("Radius: " + circle.getRadius());
        System.out.println("Area: " + circle.calculateArea());
        System.out.println("Perimeter: " +
                           circle.calculatePerimeter());
    }
}
```

```
<terminated> Main (5) [Java Applicat
RECTANGLE
Width: 5.0
Height: 4.0
Area: 20.0
Perimeter: 18.0

CIRCLE
Radius: 3.0
Area: 28.274333882308138
Perimeter: 18.84955592153876
```

Output:

Student .java that is used for Practice 8 and 7:

```java
package Exercise7;


public class Student {

    private String name;
    private String id;
    private int age;
    private String gender;

    public Student(String name, String id, int age, String gender) {
        this.name = name;
        this.id = id;
        this.age = age;
        this.gender = gender;
    }

    public void displayInfo() {
        System.out.println(
            name + " | " + id + " | " + age + " | " + gender
        );
    }
}
```

Practice 7:

CVSThread.java

```java
package Exercise7;

import java.io.BufferedReader;

public class CVSThread extends Thread {

    @Override
    public void run() {

        ArrayList<Student> students = new ArrayList<>();

        try (BufferedReader reader =
             new BufferedReader(new FileReader("students.csv"))) {

            String line;
            reader.readLine();

            while ((line = reader.readLine()) != null) {

                String[] data = line.split(",");

                if (data.length == 4) {

                    String name = data[0].trim();
                    String id = data[1].trim();
                    int age = Integer.parseInt(data[2].trim());
                    String gender = data[3].trim();

                    Student student =
                        new Student(name, id, age, gender);

                    students.add(student);
                }
            }

            System.out.println("Student List:");

            for (Student student : students) {
                student.displayInfo();
            }

        } catch (IOException e) {
            System.out.println("Error reading CSV: " + e.getMessage());
        } catch (NumberFormatException e) {
            System.out.println("Invalid student age in CSV file.");
        }
    }
}
```

PrimeThread.java

```java
package Exercise7;

public class PrimeThread extends Thread {

    @Override
    public void run() {

        System.out.println("Prime numbers from 1 to 100:");

        for (int number = 2; number <= 100; number++) {

            boolean isPrime = true;

            for (int i = 2; i * i <= number; i++) {

                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }

            if (isPrime) {
                System.out.print(number + " ");
            }
        }

        System.out.println();
    }
}
```

SumThread.java

```java
package Exercise7;

public class SumThread extends Thread {

    private int n;

    public SumThread(int n) {
        this.n = n;
    }

    @Override
    public void run() {

        int sum = 0;

        for (int i = 1; i <= n; i++) {
            sum += i;
        }

        System.out.println("Sum from 1 to " + n + " = " + sum);
    }
}
```

Main:

Main.java Rectangle.java Circle.java

```java
package Exercise7;


public class Main {
    public static void main(String[] args) {

        SumThread thread1 = new SumThread(100);
        PrimeThread thread2 = new PrimeThread();
        CVSThread thread3 = new CVSThread();
        thread1.start();
        thread2.start();
        thread3.start();
    }
}

```

Output:

```
<terminated> Main (6) [Java Application] C:\Program Files\Java\jdk-27\bin
Prime numbers from 1 to 100:
Student List:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
Sum from 1 to 100 = 5050
"John | 1001 | 20 | Male"
"Mary | 1002 | 21 | Female"
"David | 1003 | 22 | Male"
"Alice | 1004 | 19 | Female"
"Peter | 1005 | 23 | Male"
```

Practice 8

PrimeTask.java

Main.java Main.java Student.java

```java
package Exercise8;


public class PrimeTask implements Runnable {

    @Override
    public void run() {

        System.out.println("Prime numbers from 1 to 100:");

        for (int number = 2; number <= 100; number++) {

            boolean isPrime = true;

            for (int i = 2; i * i <= number; i++) {

                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }

            if (isPrime) {
                System.out.print(number + " ");
            }
        }

        System.out.println();
    }
}

```

CVSTask.java

```java
package Exercise8;


import java.io.BufferedReader;

public class CSVTask implements Runnable {

    @Override
    public void run() {

        ArrayList<Student> students = new ArrayList<>();

        try (BufferedReader reader =
             new BufferedReader(new FileReader("students.csv"))) {

            String line;

            reader.readLine();

            while ((line = reader.readLine()) != null) {

                String[] data = line.split(",");

                if (data.length == 4) {

                    String name = data[0].trim();
                    String id = data[1].trim();
                    int age = Integer.parseInt(data[2].trim());
                    String gender = data[3].trim();

                    students.add(
                        new Student(name, id, age, gender)
                    );
                }
            }

            System.out.println("Student List:");

            for (Student student : students) {
                student.displayInfo();
            }

        } catch (IOException e) {
            System.out.println("Error reading CSV: " + e.getMessage());
        } catch (NumberFormatException e) {
            System.out.println("Invalid student age in CSV file.");
        }
    }
}
```

SumTask.java

```java
package Exercise8;


public class SumTask implements Runnable {

    private int n;

    public SumTask(int n) {
        this.n = n;
    }

    @Override
    public void run() {

        int sum = 0;

        for (int i = 1; i <= n; i++) {
            sum += i;
        }

        System.out.println("Sum from 1 to " + n + " = " + sum);
    }
}
```

Main:

```java
package Exercise8;


public class Main {
    public static void main(String[] args) {

        SumTask sumTask = new SumTask(100);
        PrimeTask primeTask = new PrimeTask();
        CSVTask csvTask = new CSVTask();

        Thread thread1 = new Thread(sumTask);
        Thread thread2 = new Thread(primeTask);
        Thread thread3 = new Thread(csvTask);

        thread1.start();
        thread2.start();
        thread3.start();
    }
}

```

Output:

```
<terminated> Main (7) [Java Application] C:\Program Files\Java\jdk-27\bin\javaw.e
Prime numbers from 1 to 100:
Student List:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 Sum from 1 to 100 = 5050
71 73 79 83 89 97
"John | 1001 | 20 | Male"
"Mary | 1002 | 21 | Female"
"David | 1003 | 22 | Male"
"Alice | 1004 | 19 | Female"
"Peter | 1005 | 23 | Male"
```

Practice9

Exercise9.java

```java
package Exercise9;


import java.util.Arrays;

public class Exercise9 {
    public static void main(String[] args) {

        List<Integer> numbers =
            Arrays.asList(5, 2, 9, -1, 3, -8, 20, -50);

        System.out.println("Cube of each number:");

        numbers.stream()
               .map(n -> n * n * n)
               .forEach(n -> System.out.println(n));

        System.out.println("\nPositive numbers:");

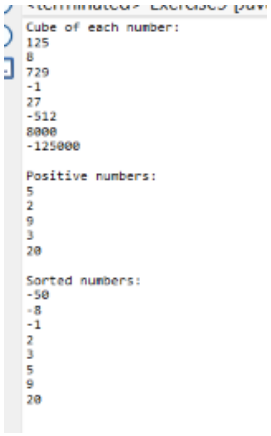
        numbers.stream()
               .filter(n -> n > 0)
               .forEach(n -> System.out.println(n));

        System.out.println("\nSorted numbers:");

        numbers.stream()
               .sorted()
               .forEach(n -> System.out.println(n));
    }
}

```

Output:



```
Cube of each number:
125
8
729
-1
27
-512
8000
-125000

Positive numbers:
5
2
9
3
20

Sorted numbers:
-50
-8
-1
2
3
5
9
20
```

Practice10:

Exercise10.java

```java
package Exercise10;


import java.io.BufferedReader;

public class Exercise10 {
    public static void main(String[] args) {

        ArrayList<String> names = new ArrayList<>();


        try (BufferedReader reader =
             new BufferedReader(new FileReader("students.csv"))) {

            String line;


            reader.readLine();

            while ((line = reader.readLine()) != null) {
                names.add(line.trim());
            }

        } catch (IOException e) {
            System.out.println("Error reading CSV: " + e.getMessage());
            return;
        }


        System.out.println("Student names in A-Z order:");

        names.stream()
             .filter(name -> !name.isEmpty())
             .sorted(String.CASE_INSENSITIVE_ORDER)
             .forEach(name -> System.out.println(name));
    }
}
```

Output:

Practice 11

Exercise11.java

```java
package Exercise11;
import java.util.Arrays;
import java.util.stream.IntStream;
public class Exercise11 {
    public static void main(String[] args) {
        int[] numbers = {5, 2, 9, 1, 3, 8};
        int max = IntStream.of(numbers)
                           .max()
                           .orElseThrow();
        int min = IntStream.of(numbers)
                           .min()
                           .orElseThrow();
        System.out.println("Numbers: " + Arrays.toString(numbers));
        System.out.println("Maximum number: " + max);
        System.out.println("Minimum number: " + min);
    }
}

```

Output

